

STAT 420/812

Computational Statistics

Lecture 00

Why do we need computer for Statistics?

- manipulate numbers. (calculation)

1, 1 + 2, 1 + ... + 100.1 ←

$$S^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{(n-1)}$$

overflow, underflow, rounding error.

- Simulation with random numbers.
Monte Carlo

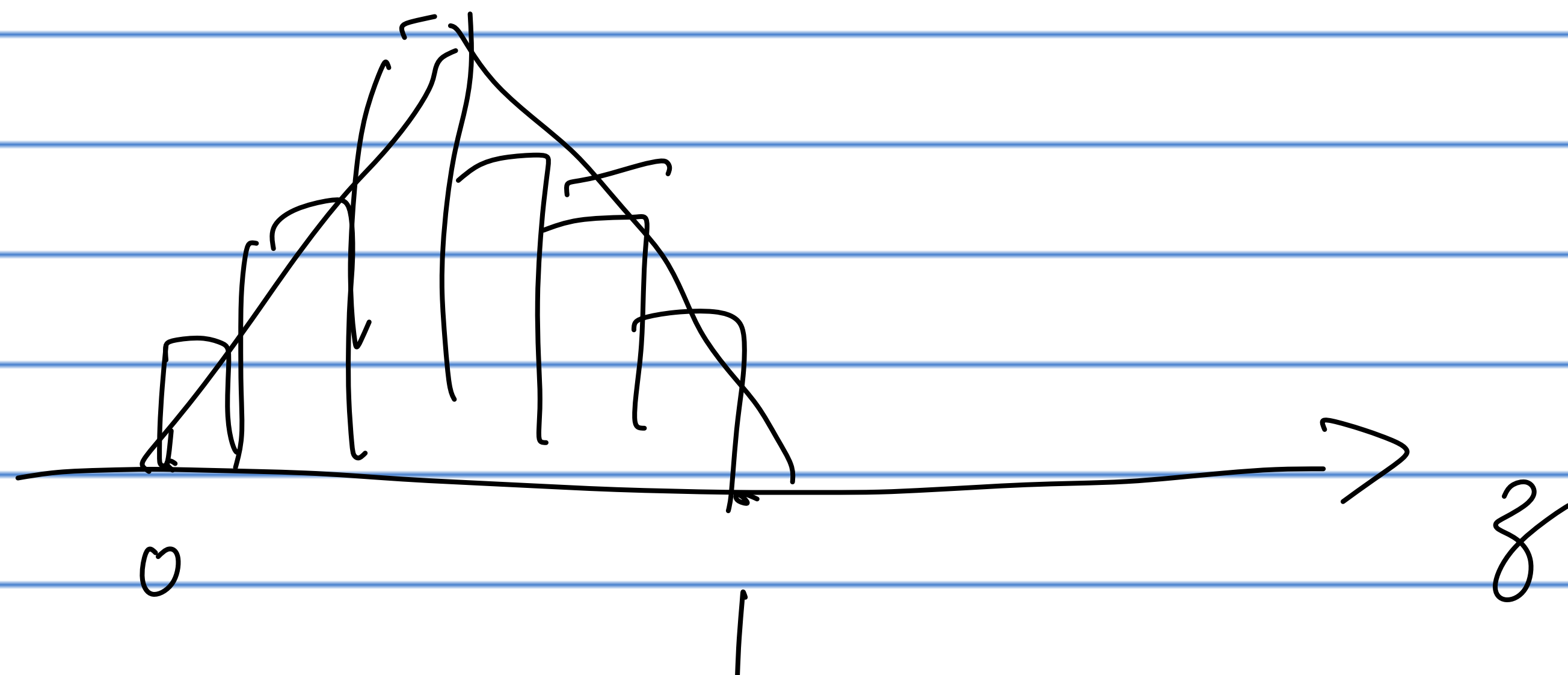
$$X_1 \sim \text{Unif}([0, 1])$$

$$X_2 \sim \text{Unif}([0, 1])$$

$$X_1 + X_2 \sim \dots$$

$$\sum X_i, \quad \sum (X_i - \bar{X})^2, \quad T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

x_1	· - - - - -	x_n	$\sim \text{unif.}$
y_1	· - - - - -	y_n	$\sim \text{unif.}$
$\downarrow$		$\downarrow$	
z_1	· · · · ·	z_n	



Particular y

$$\hat{\mu}, \hat{\sigma}^2, \hat{\beta}$$

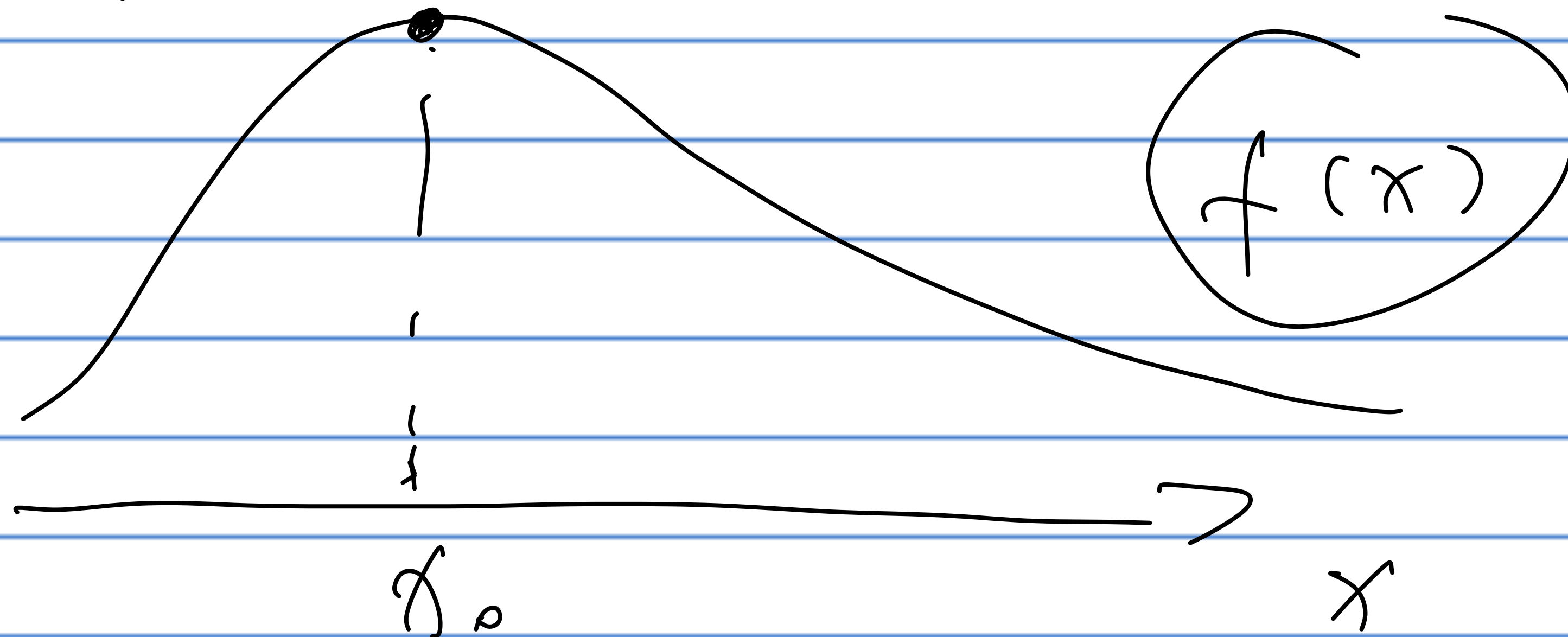
$$\hat{\mu}(x_1, \dots, x_n)$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$$\hat{\mu}^{(1)}$$

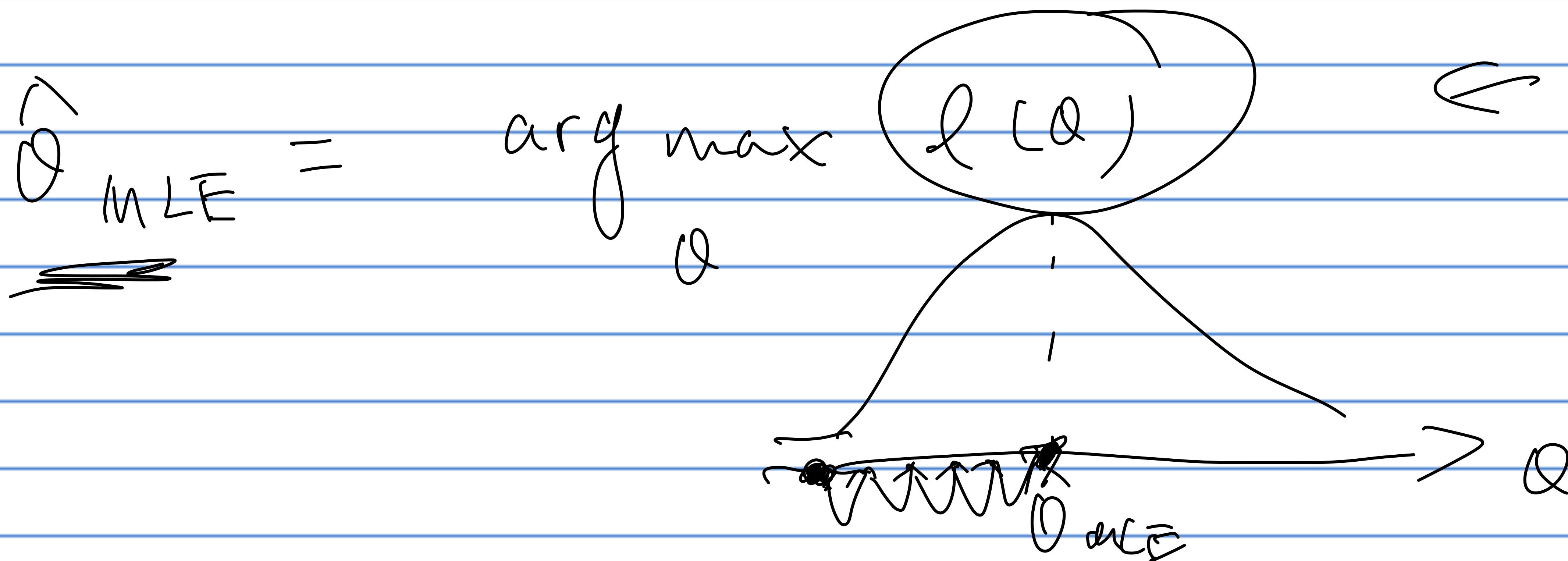
$$\hat{\mu}^{(2)}$$

- Optimization



$$x_0 = \arg \max_x f(x)$$

$$\begin{aligned}
 \ell(\theta) &= \log \left(f(\underline{x_1}, \dots, \underline{x_n} | \theta) \right) \\
 &= \log \left(\prod_{i=1}^n f(\underline{x_i} | \theta) \right) \quad \leftarrow
 \end{aligned}$$



- Advanced Simulation & Integral



$f(x)$

$p(0|x)$

$f(x|\theta) \cdot \pi(\theta)$

$E(x)$

$V(x)$

$x =$

$\frac{f(x|\theta) \cdot \pi(\theta)}{\int f(x|\theta) \pi(\theta) d\theta}$

Numerical Quadrature

